

SUMMARY

Computer Science graduate and Systems-driven Full-stack engineer, focused on choosing the right tool for each problem. Proven ability to build scalable web applications and APIs using SQL and NoSQL databases. Skilled at decomposing complex problems to deliver maintainable, high-quality code. Leverages AI for design validation, requirement refinement, and accelerated development. Seeking a collaborative team committed to deliberate architecture and long-term impact.

PROJECTS

TaskFlow | [TaskFlow Repository](#) | [Try TaskFlow](#)

- ❖ Architected MongoDB schemas with compound/text indexes and lean aggregation pipelines, eliminating N+1 queries and achieving $O(\log n)$ lookups for project-board hierarchies.
- ❖ Implemented tiered Redis caching (15-min boards, 2-min tasks) with intelligent invalidation; reduced DB load by 60% and maintained sub-50ms response times at scale.
- ❖ Built production-grade APIs with request timing middleware, selective field population, and server-side pagination - cutting payloads by 70% and enabling horizontal scaling.

WhatsApp Product Verification (WPV) | [WPV Demo](#)

- ❖ Developed a Node.js/TypeScript anti-counterfeiting platform processing millions of product verifications via WhatsApp, implementing enterprise-grade security (JWT authentication, HMAC-SHA256 webhook validation, row-level database locking).
- ❖ Eliminated race-condition fraud using PostgreSQL locking (FOR UPDATE) and achieved sub-24ms p95 response times through indexing, query optimization, and connection pooling.
- ❖ Reduced API latency by 92% (from 2,000ms industry standard to 24ms) while load-testing to 200 concurrent users with 100% uptime and zero failed requests.

Fitness Tracker | [Fitness Tracker Repository](#) | [Fitness Tracker Demo](#)

- ❖ Architected a fitness platform with decoupled microservices (React, Node.js, Python Flask) enabling workout, nutrition, and mental health tracking with ML-powered recommendations.
- ❖ Engineered multi-layer LRU caching with bounded 100-entry Flask cache and 4-minute Node.js TTL, achieving 70% cache hit rates and eliminating cold-start timeouts via async model loading on serverless infrastructure.
- ❖ Reduced AI recommendation latency by 60% through batch API parallelization and HTTP keep-alive connection pooling, delivering sub-500ms p95 response times under concurrent load with 100% service uptime.

EDUCATION

Bachelor of Science in Computer Science | Chuka University | Graduated in Nov 2025

- ❖ **Relevant Coursework:** Software Engineering, User Interface Design, Distributed Systems, Decision Support Systems, Component Based Architecture, Software Project Management, Artificial Intelligence, Data Structures.

TECHNICAL STACK

- ❖ **Frontend:** React (Vite), Next.js (App Router), TypeScript, JavaScript, Vue, Tailwind CSS, TanStack Query, Material UI, ShadcnUi, Flutter.
- ❖ **Backend & Databases:** Node.js, Express.js, Winston, Python, Fast API, Laravel, Flask, PostgreSQL, MongoDB, Redis.
- ❖ **DevOps & Tools:** GitHub Actions (CI/CD), Docker, Jest, Playwright, Vercel, Sentry, Git, GitHub, Render, Resend, Linux, CLI, Supertest, k6, Ngrok.
- ❖ **Core Competencies:** Context Engineering, Systems Thinking, AI-Augmented Development, Adaptability, Proactive Learning, Problem Solving, Abstraction & Pattern Recognition, Teamwork, Time Management.

CERTIFICATION

- ❖ Docker Bootcamp for ML, GenAI and Agentic AI.
- ❖ Prompt Engineering with Amazon Q Developer.
- ❖ Cybersecurity & Emerging Technologies Awareness.